
Adaptive Instance-Level Reweighting for Imbalanced Classification: A Comparative Study on Credit Card Fraud and Mammography Datasets

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Abstract

Class imbalance is a major challenge in machine learning where minority-class samples are significantly underrepresented compared to majority-class samples. Traditional machine learning algorithms often become biased toward the majority class and fail to correctly identify rare but important events. In this work, we implement and experimentally evaluate an adaptive instance-level reweighting strategy proposed in a recent research paper for handling imbalanced classification problems. The implemented method dynamically updates sampling probabilities based on the learning difficulty of individual training samples.

The proposed approach is evaluated on two real-world imbalanced datasets: the Credit Card Fraud Detection dataset and the Mammography dataset from the UCI Machine Learning Repository. Logistic Regression, Support Vector Machine (SVM), and Random Forest (RF) are used as baseline machine learning algorithms for comparison. In addition to reproducing the original proposed method, two extensions based on weighted-loss optimization and ensemble learning are also explored.

Experimental results demonstrate that the proposed adaptive reweighting framework achieves competitive performance relative to traditional machine learning baselines, particularly in terms of F1-score and Precision-Recall AUC (PR-AUC), which are more appropriate metrics for highly imbalanced datasets. The weighted-loss extension performs particularly well under extreme imbalance conditions, while the ensemble extension improves robustness and generalization on moderately imbalanced datasets.

GitHub Repository: DA5400 Final Project Repository

Video Presentation: Project Presentation Video

1 Motivation of the Problem

Class imbalance is a common challenge in machine learning where one class contains significantly fewer samples than the other classes. In such situations, machine learning models often become biased toward the majority class because minimizing overall classification error mainly favors majority-class predictions. As a result, minority-class samples are frequently misclassified even though they are often the most important instances in real-world applications.

Imbalanced classification problems are widely seen in areas such as fraud detection, medical diagnosis, anomaly detection, and fault prediction. In the Credit Card Fraud Detection dataset, fraudulent transactions represent only a very small portion of the total transactions, creating an extremely imbalanced distribution. Similarly, in the Mammography dataset, abnormal medical samples are much fewer than normal samples. Incorrectly classifying such minority-class samples may lead to serious financial or medical consequences. Therefore, improving minority-class detection while maintaining overall model performance is an important problem in imbalanced learning.

Traditional imbalance-handling methods such as oversampling, undersampling, SMOTE, and cost-sensitive learning mainly perform balancing at the class level. However, these approaches assume that all samples within the same class are equally important during training. In practice, some samples are more difficult to learn than others, even within the majority class. Reducing the importance of the entire majority class may therefore suppress informative or difficult samples and negatively affect model performance (1).

The research paper followed in this project, “*A Re-Balancing Strategy for Class-Imbalanced Classification Based on Instance Difficulty*”, addresses this limitation using an adaptive instance-level reweighting strategy (1). Instead of assigning weights only at the class level, the proposed framework dynamically estimates the learning difficulty of individual training samples during optimization. Samples that remain difficult to classify receive higher sampling probabilities in later training iterations, allowing the model to focus more on difficult and informative instances without explicitly modifying the original dataset distribution.

To evaluate the effectiveness of the proposed framework, experiments were conducted on two real-world imbalanced datasets:

- Credit Card Fraud Detection Dataset
- Mammography Dataset

The proposed framework was compared with three standard machine learning baseline models:

- Logistic Regression
- Support Vector Machine (SVM)
- Random Forest (RF)

In addition to reproducing the original adaptive reweighting framework, two additional extensions were explored:

- Weighted-loss extension
- Ensemble learning extension

The weighted-loss extension combines adaptive sampling with class-weighted optimization, while the ensemble extension improves robustness using prediction averaging from multiple independently trained models.

Experimental results show that the proposed adaptive instance-level reweighting framework achieves competitive performance compared to conventional baseline models, particularly for minority-class detection metrics such as F1-score and Precision-Recall AUC (PR-AUC).

The main contributions of this project are summarized as follows:

- Implementation and experimental evaluation of the adaptive instance-level reweighting strategy proposed in the research paper.
- Comparative analysis on two real-world imbalanced datasets with different imbalance characteristics.
- Performance comparison with standard machine learning baseline models including Logistic Regression, SVM, and Random Forest.
- Investigation of weighted-loss and ensemble-based extensions to improve minority-class detection and model robustness.
- Comprehensive evaluation using imbalance-sensitive metrics including Precision, Recall, F1-score, ROC-AUC, and PR-AUC.

2 Problem Statement and Methodology

2.1 Problem Formulation

Consider a binary classification dataset:

$$D = \{(x_i, y_i)\}_{i=1}^N \quad (1)$$

where $x_i \in \mathbb{R}^d$ represents the feature vector, $y_i \in \{0, 1\}$ denotes the class label, and N is the total number of samples.

In imbalanced classification problems, the number of majority-class samples is much larger than the number of minority-class samples. The imbalance ratio is defined as:

$$IR = \frac{N_{majority}}{N_{minority}} \quad (2)$$

A large imbalance ratio causes machine learning models to become biased toward the majority class, resulting in poor minority-class prediction performance. This issue is especially important in applications such as fraud detection and medical diagnosis, where minority-class samples represent critical events.

The main objective of this work is to improve minority-class detection using an adaptive instance-level reweighting strategy based on sample learning difficulty.

2.2 Adaptive Instance-Level Reweighting

The implemented method is based on the research paper “A Re-Balancing Strategy for Class-Imbalanced Classification Based on Instance Difficulty” (1). Unlike traditional class-level balancing methods, the adopted approach dynamically updates sampling importance at the instance level according to the learning behavior of individual samples.

During training, the difficulty of each sample is estimated using:

- Upward Difficulty (DU)
- Downward Learning (DL)

The difficulty score for the i^{th} sample is computed as:

$$Difficulty_i = \frac{c + DU_i}{c + DL_i} \quad (3)$$

where c is a stabilization constant used to avoid numerical instability.

Samples with larger difficulty scores are assigned higher sampling importance during the next training iteration. The adaptive sampling probability is computed as:

$$w_i = \frac{Difficulty_i}{\sum_{j=1}^N Difficulty_j} \quad (4)$$

As a result, difficult samples are selected more frequently during training, allowing the model to focus more on informative instances without explicitly modifying the original dataset distribution using oversampling or undersampling techniques.

2.3 Neural Network Architecture

A fully connected feedforward neural network was used to implement the adaptive reweighting framework. The network consists of multiple fully connected layers with ReLU activation functions and dropout regularization to improve nonlinear learning capability and reduce overfitting during training.

The detailed network architecture used in this work is summarized in Table 1.

Table 1: Neural network architecture used in the proposed framework

Layer	Configuration
Input Layer	Feature Input
Hidden Layer 1	Fully Connected (128 neurons) + ReLU + Dropout
Hidden Layer 2	Fully Connected (64 neurons) + ReLU + Dropout
Hidden Layer 3	Fully Connected (32 neurons) + ReLU
Output Layer	Fully Connected (2 neurons)

2.4 Framework Extensions

In addition to reproducing the original adaptive reweighting framework, two additional extensions were explored in this work.

The first extension combines adaptive sampling with class-weighted cross-entropy loss to further emphasize minority-class learning during training. The weighted loss function is expressed as:

$$L = - \sum_{i=1}^N w_{y_i} \log(p_{y_i}) \quad (5)$$

where w_{y_i} denotes the class weight and p_{y_i} represents the predicted probability of the true class.

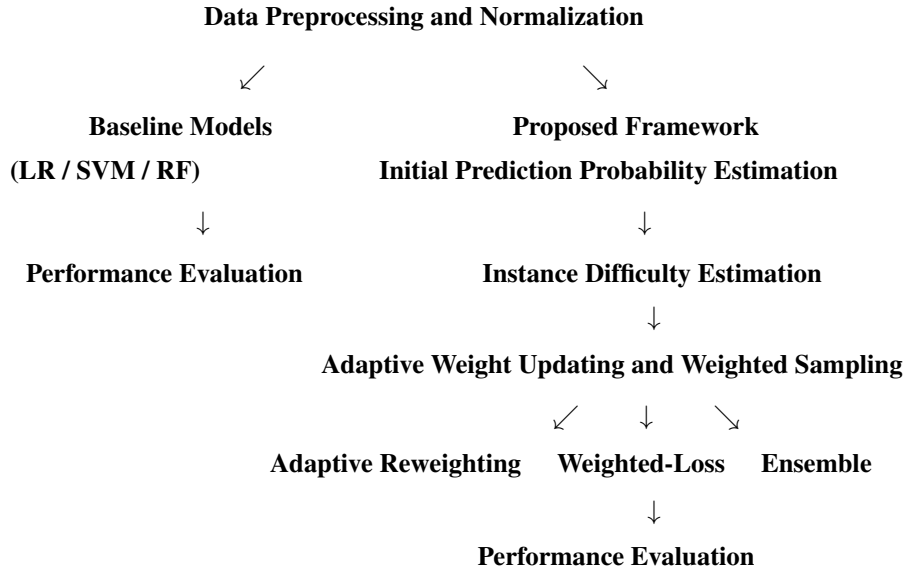
The second extension uses ensemble learning to improve robustness and reduce prediction variance. Multiple adaptive reweighting models were trained independently, and the final prediction probabilities were obtained by averaging the outputs of all ensemble members:

$$P_{ensemble} = \frac{1}{M} \sum_{k=1}^M P_k \quad (6)$$

where M represents the number of models and P_k denotes the prediction probability of the k^{th} model.

2.5 Overall Workflow

The overall workflow of the implemented framework is summarized below:



3 Datasets

To evaluate the effectiveness of the proposed adaptive reweighting framework, experiments were conducted on two real-world imbalanced classification datasets with different imbalance characteristics. The selected datasets represent important practical applications in fraud detection and medical diagnosis.

Table 2 summarizes the key properties of the datasets used in this work.

Table 2: Summary of datasets used in this work

Dataset	Samples	Features	Minority Samples	Imbalance Ratio
Credit Card Fraud	284,807	30	492	577:1
Mammography	11,183	6	260	42:1

3.1 Credit Card Fraud Detection Dataset

The Credit Card Fraud Detection dataset contains financial transaction records collected from European cardholders. The objective is to classify transactions as either normal or fraudulent. The dataset contains 284,807 samples with 30 input features, of which only 492 correspond to fraud cases, resulting in an extreme imbalance ratio of approximately 577:1.

Most input features are already anonymized using PCA transformation for privacy preservation, while two additional features correspond to transaction time and transaction amount.

3.2 Mammography Dataset

The Mammography dataset was obtained from the UCI Machine Learning Repository for anomaly detection and medical diagnosis tasks. The objective is to classify samples as either normal or abnormal.

The dataset contains 11,183 samples with 6 numerical features extracted from mammographic measurements. Among the total samples, only 260 belong to the abnormal class, resulting in an imbalance ratio of approximately 42:1. Although the imbalance is less severe than the Credit Card Fraud dataset, reliable minority-class detection remains important because false negatives may lead to serious medical risks.

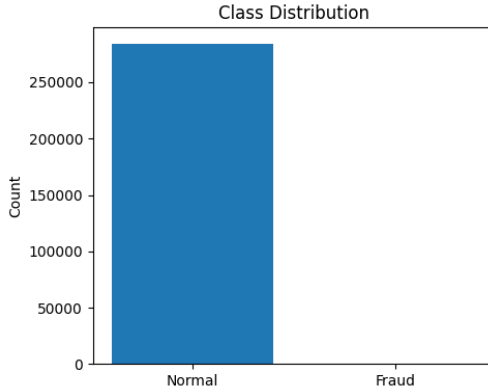


Figure 1: Class distribution of the Credit Card Fraud dataset.

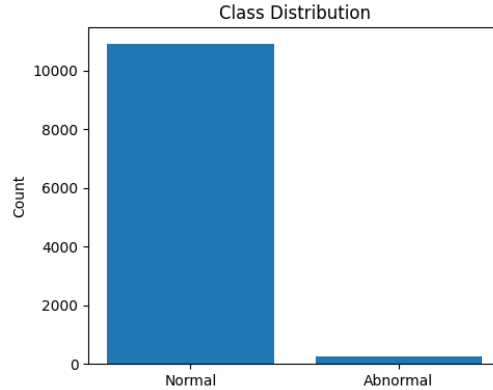


Figure 2: Class distribution of the Mammography dataset.

3.3 Data Preprocessing

Before training, feature normalization using standard scaling was applied to improve optimization stability and training performance. The datasets were further divided into training and testing sets, followed by conversion into PyTorch tensors for model implementation.

4 Experimental Setup and Baselines

The proposed adaptive reweighting framework was implemented using PyTorch, while baseline machine learning models were implemented using Scikit-learn. All experiments were conducted using CUDA GPU acceleration.

4.1 Baseline Models

To evaluate the effectiveness of the proposed framework, three standard machine learning algorithms were used as baseline models:

- **Logistic Regression (LR)** – linear baseline model for binary classification.
- **Support Vector Machine (SVM)** – margin-based classifier effective for high-dimensional data.
- **Random Forest (RF)** – ensemble-based decision tree classifier with strong nonlinear learning capability.

4.2 Proposed Variants

Three variants of the adaptive reweighting framework were evaluated:

- **Adaptive Reweighting** – original instance-level reweighting framework proposed in the research paper.
- **Weighted-Loss Extension** – adaptive reweighting combined with class-weighted cross-entropy optimization.
- **Ensemble Extension** – prediction averaging from multiple independently trained adaptive reweighting models.

4.3 Training Configuration

The neural network models were trained using the Adam optimizer with mini-batch learning. Dropout regularization and adaptive instance-level sampling were incorporated to improve generalization and emphasize difficult training samples during optimization.

The main training hyperparameters are summarized in Table 3.

Table 3: Training hyperparameters used in the experiments

Parameter	Value
Optimizer	Adam
Learning Rate	0.001
Batch Size	512
Epochs	100
Dropout Rate	0.3 / 0.2
Loss Function	Cross-Entropy Loss
Device	CUDA GPU

4.4 Training Procedure

During training, instance-level sampling weights were updated after each epoch according to sample learning difficulty. Prediction probabilities were used to compute normalized difficulty scores, which guided weighted random sampling in subsequent iterations. For the weighted-loss extension, class-weighted cross-entropy loss was combined with adaptive sampling, while the ensemble extension averaged predictions from multiple independently trained adaptive reweighting models.

5 Evaluation Metrics

Accuracy alone is not sufficient for evaluating models on imbalanced datasets because a model may achieve high accuracy by predicting only the majority class. Therefore, multiple evaluation metrics were used in this work to properly evaluate minority-class prediction performance.

Table 4 summarizes the evaluation metrics used in this project.

Table 4: Evaluation metrics used in the experiments

Metric	Formula	Description
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Measures overall classification performance.
Precision	$\frac{TP}{TP+FP}$	Measures how many predicted positive samples are actually positive.
Recall	$\frac{TP}{TP+FN}$	Measures how many actual positive samples are correctly detected.
F1-Score	$\frac{2 \times Precision \times Recall}{Precision + Recall}$	Provides a balanced evaluation using both Precision and Recall.
ROC-AUC	–	Measures the ability of the model to distinguish between classes across different thresholds.
PR-AUC	–	Evaluates the Precision-Recall trade-off and is more informative for highly imbalanced datasets.

In addition to numerical metrics, confusion matrices were also used to analyze false positives, false negatives, and correctly classified samples for different models.

6 Results and Analysis

This section presents the experimental results obtained on the Credit Card Fraud and Mammography datasets. The proposed adaptive reweighting framework and its extensions were compared against traditional machine learning baseline models using imbalance-sensitive evaluation metrics.

6.1 Results on Credit Card Fraud Dataset

The Credit Card Fraud dataset represents an extremely imbalanced classification problem with an imbalance ratio of approximately 577:1. The quantitative performance comparison of all evaluated methods is presented in Table 5.

Table 5: Performance comparison on the Credit Card Fraud dataset

Method	Accuracy	Precision	Recall	F1-score	ROC-AUC	PR-AUC
Logistic Regression	0.99917	0.7895	0.7095	0.7473	0.9566	0.7088
SVM	0.99945	0.9244	0.7432	0.8240	0.9419	0.8114
Random Forest	0.99958	0.9667	0.7838	0.8657	0.9365	0.8274
Proposed Adaptive Reweighting	0.99949	0.9333	0.7568	0.8358	0.8994	0.8269
Weighted-Loss Extension	0.99947	0.9120	0.7703	0.8352	0.8918	0.8612
Ensemble Extension	0.99945	0.9244	0.7432	0.8240	0.9400	0.8111

Among the baseline models, Random Forest achieved the best overall performance among the baseline models with the highest F1-score of 0.8657 along with strong Precision and Recall values. This shows that Random Forest handled the severe class imbalance more effectively than Logistic Regression and SVM on the Credit Card Fraud dataset.

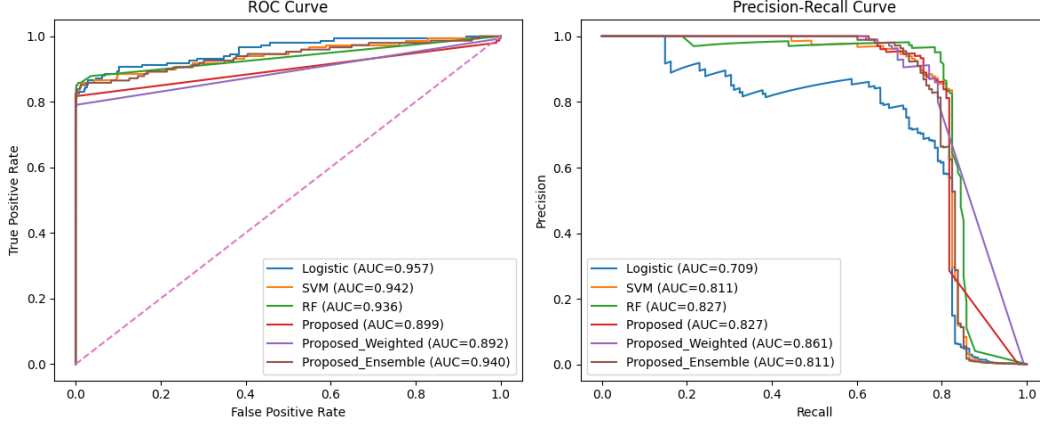


Figure 3: ROC and Precision-Recall curves for the Credit Card Fraud dataset.

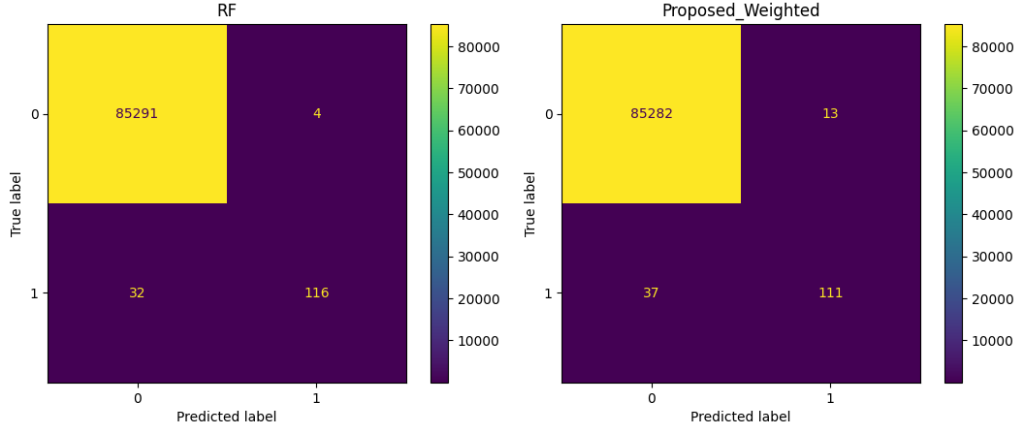


Figure 4: Confusion matrices for Random Forest and Weighted-Loss Extension on the Credit Card Fraud dataset.

The proposed adaptive reweighting framework also achieved competitive minority-class detection performance. Compared to Logistic Regression, the proposed framework improved Precision, Recall, F1-score, and PR-AUC, indicating that adaptive instance-level sampling helped the model focus more effectively on difficult fraud samples during training.

Among all evaluated methods, the Weighted-Loss Extension achieved the highest PR-AUC score of 0.8612. Since PR-AUC directly reflects minority-class prediction quality, this result suggests that combining adaptive sampling with class-weighted optimization improved fraud detection performance under extreme imbalance conditions.

Figure 3 presents the ROC and Precision-Recall curves for all evaluated methods on the Credit Card Fraud dataset. The proposed methods maintained stable performance across different thresholds, particularly in the Precision-Recall curves which provide more reliable evaluation for highly imbalanced datasets.

The confusion matrices shown in Figure 4 further illustrate the prediction behavior of Random Forest and the Weighted-Loss Extension. The Weighted-Loss Extension reduced false-negative predictions while maintaining high overall classification accuracy.

Figure 5 shows the variation of Precision and Recall across different classification thresholds. As the threshold increased, Precision improved while Recall gradually decreased, demonstrating the trade-off between reducing false positives and maintaining minority-class detection sensitivity.

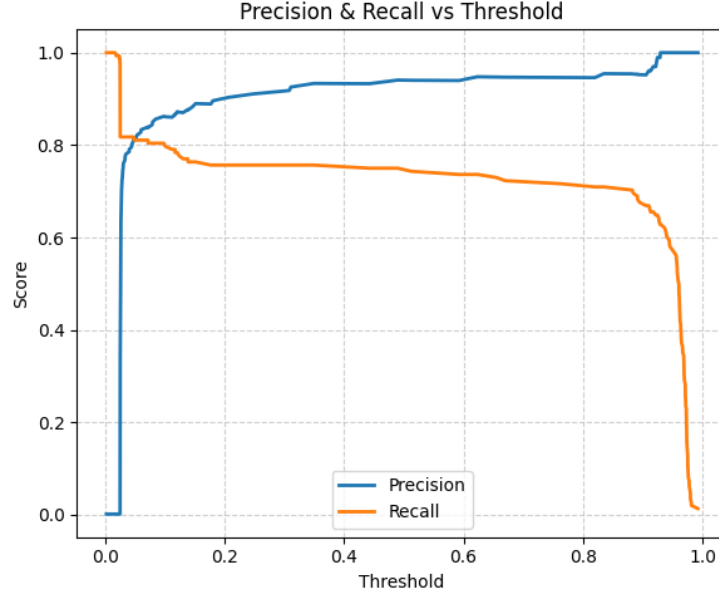


Figure 5: Precision and Recall variation across different classification thresholds for the Credit Card Fraud dataset.

6.2 Results on Mammography Dataset

The Mammography dataset represents a moderately imbalanced classification problem with an imbalance ratio of approximately 42:1. Table 6 summarizes the experimental results obtained on this dataset.

Table 6: Performance comparison on the Mammography dataset

Method	Accuracy	Precision	Recall	F1-score	ROC-AUC	PR-AUC
Logistic Regression	0.9818	0.6024	0.6410	0.6211	0.9619	0.6490
SVM	0.9848	0.6667	0.6923	0.6792	0.8999	0.6660
Random Forest	0.9884	0.7671	0.7179	0.7417	0.9637	0.7530
Proposed Adaptive Reweighting	0.9875	0.7308	0.7308	0.7308	0.9668	0.7717
Weighted-Loss Extension	0.9878	0.7229	0.7692	0.7453	0.9696	0.7656
Ensemble Extension	0.9890	0.7971	0.7051	0.7483	0.9734	0.7884

The Ensemble Extension achieved the best overall performance with the highest Accuracy (0.9890), F1-score (0.7483), ROC-AUC (0.9734), and PR-AUC (0.7884). These results indicate that ensemble learning improved robustness and generalization under moderate imbalance conditions.

The proposed adaptive reweighting framework also achieved competitive performance compared to the baseline models. In particular, the proposed methods achieved better Recall and PR-AUC values than Logistic Regression and SVM, demonstrating improved abnormal sample detection capability.

The Weighted-Loss Extension achieved higher Recall compared to Random Forest, indicating improved minority-class detection performance. Figure 6 presents the ROC and Precision-Recall curves for all evaluated models on the Mammography dataset.

The confusion matrices shown in Figure 7 demonstrate that the Ensemble Extension reduced false-negative predictions while maintaining low false-positive rates. This indicates improved robustness and minority-class prediction capability.

Figure 8 illustrates the variation of Precision and Recall across different thresholds for the proposed framework. Similar to the Credit Card Fraud dataset, increasing the threshold improves Precision

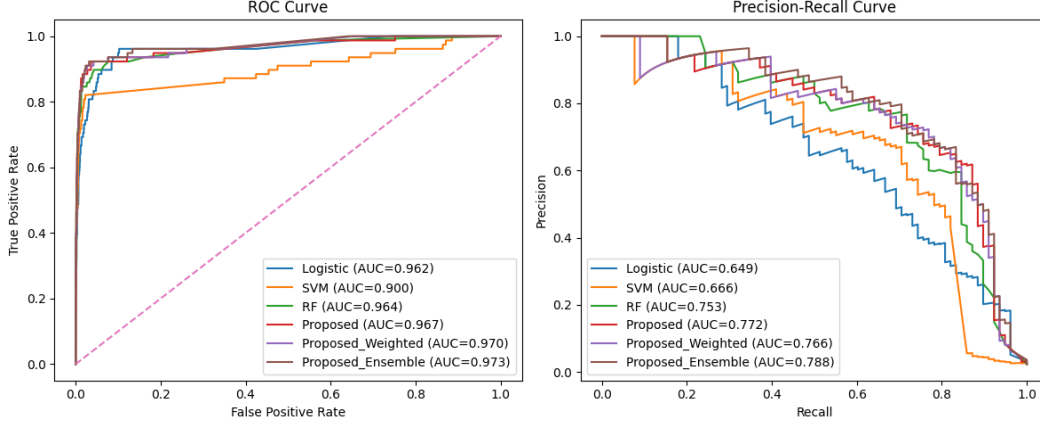


Figure 6: ROC and Precision-Recall curves for the Mammography dataset.

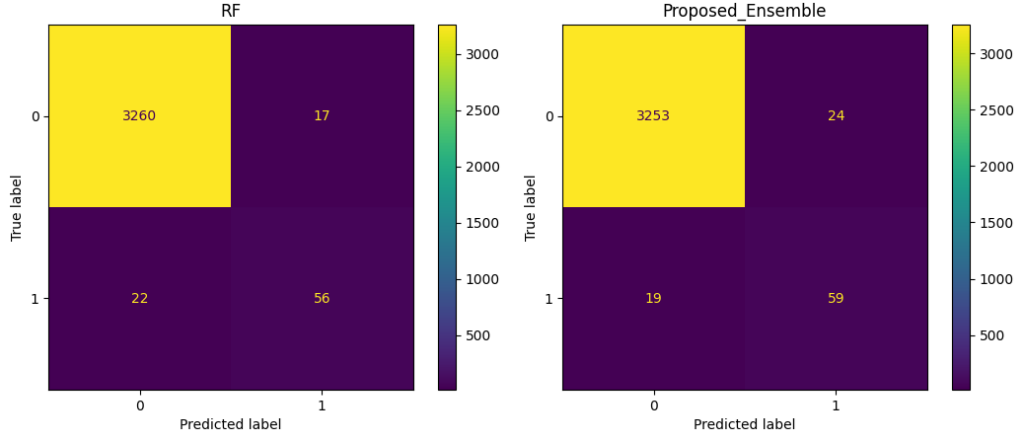


Figure 7: Confusion matrices for Random Forest and Ensemble Extension on the Mammography dataset.

while reducing Recall, highlighting the trade-off between detection sensitivity and prediction confidence.

6.3 Overall Observations

The experimental results show that adaptive instance-level weighting improved minority-class prediction performance across different imbalance conditions. The proposed methods achieved better minority-class detection capability than Logistic Regression and SVM, especially on the highly imbalanced Credit Card Fraud dataset.

The results also show that different extensions of the framework performed differently depending on the dataset characteristics. The Weighted-Loss Extension performed better under severe imbalance conditions, while the Ensemble Extension produced more stable overall performance on the Mammography dataset.

Another important observation was that Random Forest remained a strong baseline model across both datasets, especially for structured tabular data where tree-based ensemble methods effectively capture nonlinear feature relationships.

Overall, the experiments demonstrate that adaptive sampling based on instance difficulty is an effective approach for handling imbalanced classification problems.

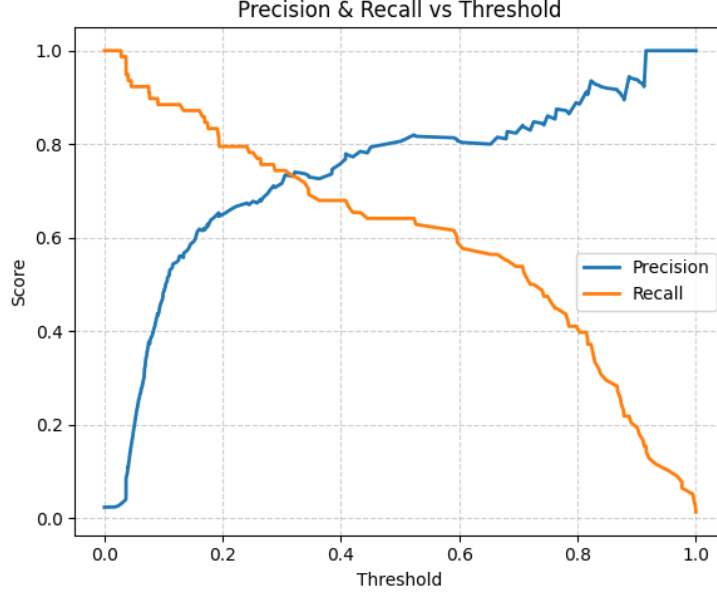


Figure 8: Precision and Recall variation across different classification thresholds for the Mammography dataset.

7 Discussion

The experimental results obtained in this project are consistent with the main claims presented in the research paper (1). The paper proposes that adaptive instance-level reweighting improves minority-class learning by dynamically assigning higher importance to difficult training samples during optimization. The results obtained from our implementation support this observation across both datasets.

On the Credit Card Fraud dataset, which contains extremely severe imbalance, the proposed adaptive framework achieved stronger minority-class detection performance compared to Logistic Regression and SVM. In particular, the Weighted-Loss Extension achieved the highest PR-AUC score of 0.8612 among all evaluated methods. Since PR-AUC directly reflects minority-class prediction quality under highly imbalanced conditions, this result indicates that combining adaptive sampling with weighted optimization improves fraud detection capability.

The Precision-Recall curves and confusion matrices further verified the claims of the paper. The proposed methods reduced false-negative predictions for fraudulent transactions while maintaining stable Precision and Recall values. This suggests that difficult minority-class samples received greater importance during training through adaptive sampling updates.

The paper also claims that imbalance handling can be improved without explicitly modifying the original dataset distribution using oversampling or undersampling techniques (1). The implementation used in this project followed the same principle by dynamically updating sampling probabilities according to instance difficulty during training. The competitive performance obtained on both datasets supports the effectiveness of this adaptive weighted sampling strategy.

On the Mammography dataset, which contains moderate imbalance compared to the Credit Card Fraud dataset, the Ensemble Extension achieved the best overall performance with Accuracy of 0.9890, F1-score of 0.7483, ROC-AUC of 0.9734, and PR-AUC of 0.7884. The confusion matrices additionally showed fewer false-negative predictions compared to several baseline methods, indicating improved robustness and prediction stability.

The experiments also verified that different imbalance conditions require different optimization strategies. The Weighted-Loss Extension performed best on the extremely imbalanced Credit Card Fraud dataset, while the Ensemble Extension achieved stronger overall performance on the Mammography dataset.

Another important observation from the experiments was that Random Forest remained a strong baseline model across both datasets. On the Credit Card Fraud dataset, Random Forest achieved the highest F1-score of 0.8657, showing that tree-based ensemble methods remain highly effective for structured tabular imbalance problems.

Table 7 summarizes the comparison between the claims presented in the research paper and the observations obtained from the experimental analysis.

Table 7: Comparison between claims of the research paper and experimental observations

Claim in Research Paper	Observation from Experiments
Adaptive instance-level weighting improves minority-class learning	The proposed framework improved Recall, F1-score, and PR-AUC compared to Logistic Regression and SVM.
Difficult samples should receive higher importance during training	The proposed methods reduced false-negative predictions in the confusion matrices.
Adaptive sampling improves imbalance handling without oversampling or undersampling	Competitive minority-class detection performance was achieved without modifying the original dataset distribution.
Different imbalance conditions require different optimization strategies	Weighted-Loss Extension performed best on the Credit Card Fraud dataset, while Ensemble Extension achieved better overall performance on the Mammography dataset.

Overall, the experimental analysis performed in this project validates the central findings of the research paper and demonstrates that adaptive instance-level sampling based on learning difficulty is an effective strategy for handling imbalanced classification problems.

8 Limitations

Few limitations were observed during the implementation and experimentation of the proposed adaptive reweighting framework.

The adaptive sampling process increased training complexity because instance-level difficulty scores and sampling probabilities were updated after every training epoch. As a result, the proposed framework required longer training time compared to conventional baseline models such as Logistic Regression and Random Forest.

The performance of the framework was also sensitive to hyperparameter selection, particularly classification thresholds, weighting updates, and training configuration. Small changes in these parameters affected the Precision-Recall trade-off and minority-class detection performance.

Another important observation from the experiments was that Random Forest remained a very strong baseline model across both datasets. On the Credit Card Fraud dataset, Random Forest achieved the highest F1-score, while on the Mammography dataset the improvement obtained using the proposed framework was relatively small. This suggests that tree-based ensemble methods are highly effective for structured tabular imbalance problems.

In addition, the Ensemble Extension further increased computational cost because multiple adaptive reweighting models had to be trained independently before prediction averaging during inference.

Finally, the experiments conducted in this work were limited to binary classification datasets only. The effectiveness of adaptive instance-level reweighting for multiclass imbalance problems was not investigated.

References

- [1] S. Yu, J. Guo, R. Zhang, Y. Fan, Z. Wang, and X. Cheng, “A Re-Balancing Strategy for Class-Imbalanced Classification Based on Instance Difficulty,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 502–511.